Expt. No: **Date:**

1. PSEUDOCODE AND FLOWCHART FOR THE BIGGEST OF THREE NUMBERS

AIM:

To write pseudocode and draw flowchart for the greatest (biggest) of given three numbers.

ALGORITHM:

Step 1: Start.

Step 2: Read the values of a, b, c.

Step 3: Check if ((a > b) and (a > c)) then

3.1: print ‘A is big’.

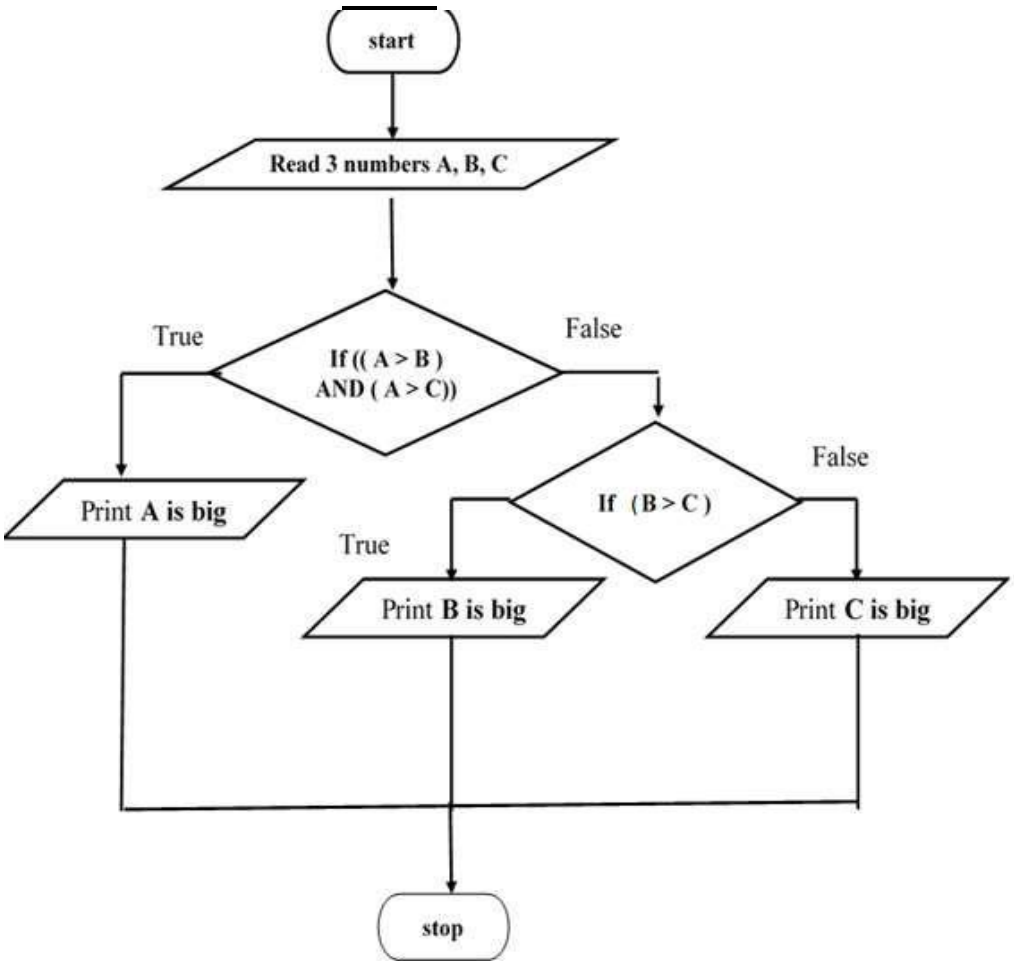
Step 4: Else if (b> c) then

4.1: print ‘B is big’.

Step 5: Else print ‘C is big’.

Step 6: Stop.

FLOWCHART:


start
Read 3 numbers A, B, C
True
If ((A > B)
AND (A > C))
False
Print A is big
If (B > C)
False
True
Print B is big
Print C is big
stop

PSEUDOCODE:

```
READ the values for a, b, c.

IF ( ( a > b ) and ( a > c ) ) then WRITE ‘A is big’.

ELSE IF if( b> c ) then WRITE ‘B is big’.

ELSE WRITE ‘C is big’.

END IF.
```

RESULT:

The pseudocode and flowchart to find the greatest number from three numbers are written and drawn successfully.

Expt. No: **Date:**

2. PSEUDOCODE AND FLOWCHART TO FIND THE SUM OF N NUMBERS

AIM:

To write pseudo-code and draw flowchart to find the sum of N numbers.

ALGORITHM:

Step 1: Start.

Step 2: Read the values of n.

Step 3: Assign sum ← 0 and initialize i← 1.

Step 4: If (i<= n) then go to step 5 else go to step 6.

Step 5: sum = sum + i.

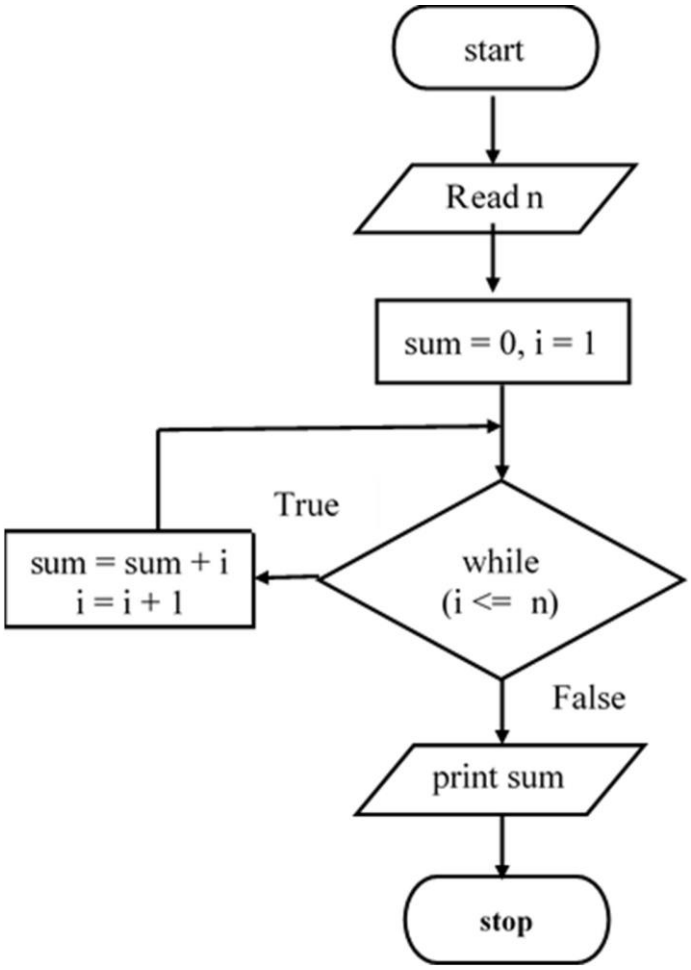
i = i + 1.

Go to step 4.

Step 6: Print sum.

Step 7: Stop.

FLOWCHART:


start
Read n
sum = 0, i = 1
True
sum = sum + i
i = i + 1
while
(i <= n)
False
print sum
stop

PSEUDOCODE:

```
READ value for n.

SET initial value for sum = 0 and i = 1.

FOR ( i<= n )

    sum = sum + i

    i = i + 1

END FOR

WRITE the result of sum
```

RESULT:

Thus the pseudocode and flowchart for sum of N numbers are written and implemented successfully.

Expt. No: **Date:**

3. PSEUDOCODE AND FLOWCHART TO COMPUTE nC_r

AIM:

To write pseudo-code and draw flowchart to compute of nCr.

We know that formula for nCr = n! / (r! * (n-r)!)

ALGORITHM:

Step 1: Start.

Step 2: Read the values of n and r.

Step 3: Call the function factorial and find value of (fact (n)).

Step 4: Call the function factorial and find value of (fact (r)).

Step 5: Call the function factorial and find value of (fact (n-r)).

Step 6: Use formula: fact(n) / ((fact(r) * fact(n-r)).

Step 7: Print the result.

Step 8: Stop.

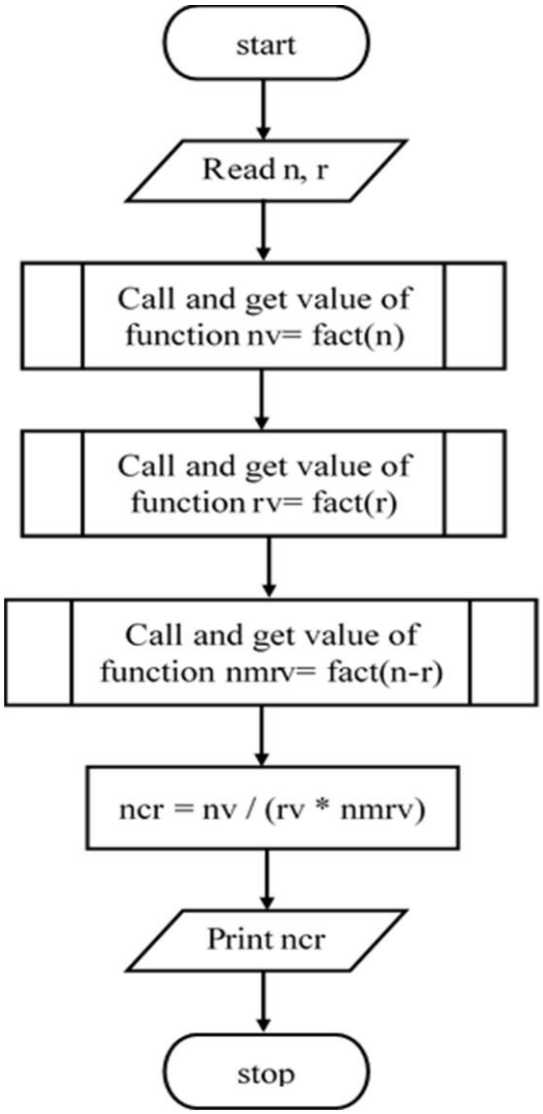
Function: fact(n).

Step 1: Start the function

Step 2: If n == 0 or n == 1 return value 1. Else go to step 3.

Step 3: Return (n * fact (n – 1))

FLOWCHART:


start
Read n, r
Call and get value of function nv= fact(n)
Call and get value of function rv= fact(r)
Call and get value of function nmrv= fact(n-r)
ncr = nv / (rv * nmrv)
Print ncr
stop

PSEUDOCODE:

READ value for n, r.

CALL THE FUNCTION fact with value n as :nv = fact(n)

CALL THE FUNCTION fact with value r as :rv = fact(r)

CALL THE FUNCTION fact with value (n-r) as :nrv = fact(n - r)

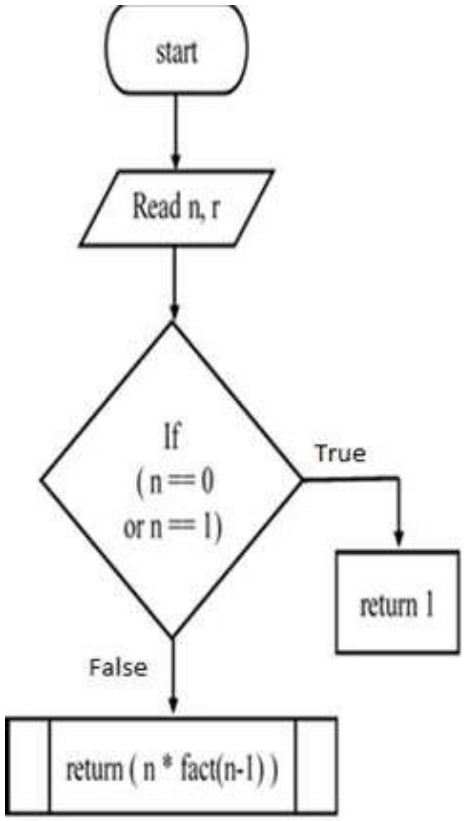
CALCULATE value using formula: fact(n) / (fact(r) * fact(n-r))

WRITE nC_r value.

Function fact(n):

IF (n == 0 or n == 1) then return value as 1

ELSE return (n * fact(n-1))


start
Read n, r
If
(n == 0
or n == 1)
True
return 1
False
return (n * fact(n-1))

RESULT:

Thus the pseudo code and flowchart for combination nC_r are written and implemented successfully.

Expt. No: **Date:**

4. PROGRAM TO READ AND PRINT DATA OF DIFFERENT TYPES

AIM:

To write a program to read and print data of different types.

ALGORITHM

Step 1: Start.

Step 2: Declare the variables of types integer, float, character, double, short int, unsigned int, long double.

Step 3: Read the values of these variables.

Step 4: Print the values of the variables.

Step 5: Stop.

PROGRAM:

```
#include <stdio.h>
#include <conio.h>
void main()
{
   int a;
   float b;
   char c;
   double d;
   short int e;
```

```
    unsigned int ui;
    long double ld;
    printf("\n Enter an Integer value: " );
    scanf("%d",&a);
    printf("\n Enter a Float value: " );
    scanf("%f",&b);
    printf("\n Enter a Character: " );
    fflush(stdin);
    scanf("%c",&c);
    printf("\n Enter a Double value: " );
    scanf("%lf",&d);
    printf("\n Enter a Short Integer value: " );
    scanf("%hi",&e);
    printf("\n Enter a Unsigned Integer value: " );
    scanf("%u",&ui);
    printf("\n Enter a Long Double value: " );
    scanf("%Lf",&ld);
    printf("\n Integer: %d \n\n Float: %f \n\n Character: %c \n\n Double: %f",a,b,c,d);
    printf("\n\n Short Integer: %hi \n\n Unsigned Integer: %u \n\n Long Double: %Lf
  ",e,ui,ld);
    getch();
  }
```

OUTPUT:

D:\BC5\BIN\datatypes.exe

```
Enter an Integer value: -33453

Enter a Float value: 123456.374

Enter a Character: G

Enter a Double value: 123456789.123

Enter a Short Integer value: 4536

Enter a Unsigned Integer value: 56

Enter a Long Double value: 123456789012345.658

Integer: -33453

Float: 123456.375000

Character: G

Double: 123456789.123000

Short Integer: 4536

Unsigned Integer: 56

Long Double: 123456789012345.658000
```

RESULT:

Reading and printing data of different types are implemented using a program.

Expt. No: **Date:**

5. PROGRAM TO ILLUSTRATE ARITHMETIC OPERATORS

AIM:

To write a program to implement arithmetic operators in C

ALGORTIHM:

Step1: Start the program

Step2: Declare the integer variables a,b and c.

Step3: Declare a float variable d.

Step4: Read the values of a and b.

Step5: Compute c = a + b

Step6: Print the result of the addition.

Step7: Compute c = a – b.

Step8: Print the result of the subtraction.

Step9: Compute c = a * b.

Step10: Print the result of multiplication.

Step11: Compute d = a/b by converting integer variable a into float to get the result in float.

Step12: Print the result of the Division.

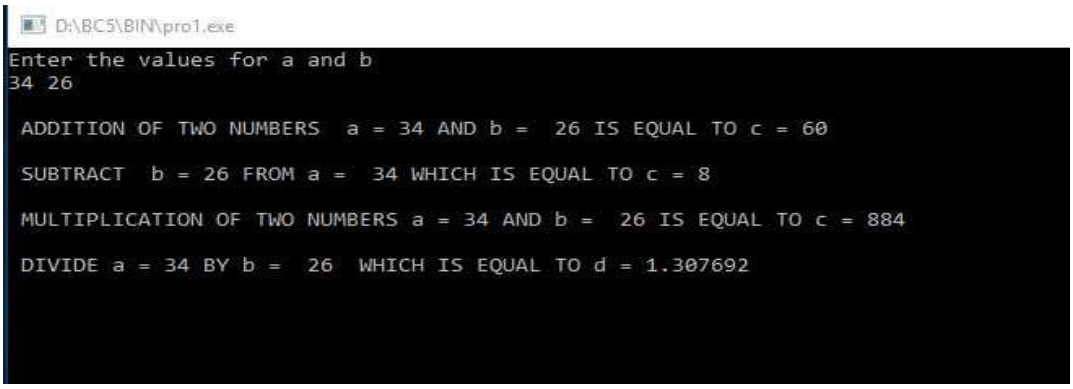
Step13: End

PROGRAM:

```
# include <stdio.h>

# include <conio.h>

void main()

{

int a, b, c;

  float d;

  /* READ THE VALUES OF TWO INTEGERS AND PERFORM ADDITION,

  SUBTRACTION AND MULTIPLICATION */

printf("Enter the values for a and b \n");

scanf(" %d %d", &a,&b);

  //ADDITION

  c = a + b;

printf("\n ADDITION OF TWO NUMBERS a = %d AND b = % d IS EQUAL TO c =
%d\n", a,b,c);

  //SUBTRACTION

  c = a - b;

printf("\n SUBTRACT b = %d FROM a = % d WHICH IS EQUAL TO c = %d\n", b,a,c);

  //MULTIPLICATION

   c = a * b;

printf("\n MULTIPLICATION OF TWO NUMBERS a = %d AND b = % d IS EQUAL TO
c = %d\n", a,b,c);
```

```
//DIVISION

d = float(a)/b;    // CONVERT INTEGER VARIABLE a TO FLOAT VARIABLE USE float(a)

printf("\n DIVIDE a = %d BY b = % d WHICH IS EQUAL TO d = %f\n", a,b,d);

getch();

}
```

OUTPUT:



RESULT:

Arithmetic operators are implemented using C program.

Expt. No: **Date:**

6. PROGRAM TO ILLUSTRATE RELATIONAL AND LOGIC OPERATORS

AIM:

To write a program to implement relational and logic operators in C

ALGORTIHM 1: RELATIONAL OPERATORS

Step1: Start the program

Step2: Declare the integer variables a and b.

Step3: Assign values to a and b.

Step4: Print the output of the comparison a is equal to b.

Step5: Print the output of the comparison a is not equal to b.

Step6: Print the output of the comparison a is greater than b.

Step7: Print the output of the comparison a is less than b.

Step8: Print the output of the comparison a is greater than or equal to b.

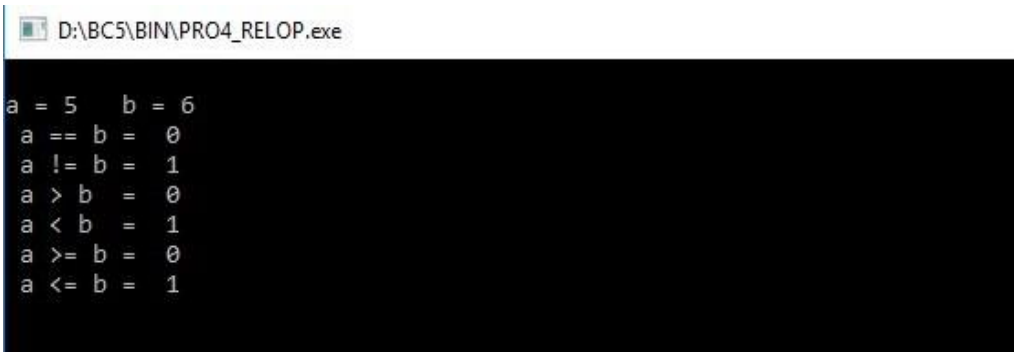
Step9: Print the output of the comparison a is less than equal to b.

Step10: End

PROGRAM 1: RELATIONAL OPERATORS

```
#include <stdio.h>
#include <conio.h>
void main()
{
int a = 5, b = 6;
printf(" \na = %d  b = %d", a, b);
printf("\n a == b = %d", a == b); // Output is 1 if a and b are equal else 0
printf("\n a != b = %d", a != b); // Output is 1 if a is not equal to b else 0
printf("\n a > b = %d", a > b); // Output is 1 if a is greater than b else 0
printf("\n a < b = %d", a < b); // Output is 1 if a is less than b else 0
printf("\n a >= b =  %d", a >= b);   // Output is 1 if a is greater than or equal to b else
0
printf("\n a <= b = %d", a <= b); // Output is 1 if a is less than or equal to b else 0
getch();
}
```

OUTPUT:



ALGORITHM 2: LOGIC OPERATOR

Step1: Start the program

Step2: Declare the integer variables a and b.

Step3: Assign values to a and b.

Step4: Print the output as 1 if a and b are non zero values else 0.

Step5: Print the output as 1 if any one of a or b is non zero value else 0.

Step6: Print the output as 1 if the comparison a is greater than b.

Step7: Print the output as 1 if a is non zero 0 if a is zero.

Step8: Print the output as 1 if a is non zero 0 if a is zero.

Step9: End

PROGRAM 2: LOGIC OPERATOR

```
#include <stdio.h>

#include <conio.h>

void main()

{

int a = 16, b = 0;

printf(" \na = %d  b = %d", a, b);

   // The output is 1 if a and b are non zerovalues, 0 otherwise

printf("\n a&&b = %d", a && b);

   //The output is 1 if any one of a or b is non zero, 0 if both are zeros

printf("\n a||b = %d", a || b);

   //The output is 1 if a is zero , 0 if a is non zero

printf("\n !a = %d", !a);

   //The output is 1 if b is zero , 0 if b is non zero

printf("\n !b = %d", !b);
getch();
}
```

OUTPUT:

RESULT:

Relational operators and Logical operators are implemented using C programs.

Expt. No: **Date:**

7. PROGRAMS TO ILLUSTRATE ASSIGNMENT, INCREMENT AND DECREMENT, SIZE OF AND BITWISE OPERATORS

AIM:

To write programs to implement assignment, increment and decrement, size of and bitwise operators in C.

ALGORITHM 1: ASSIGNMENT OPERATORS

Step1: Start.

Step2: Declare the variables of type integers and assign the values.

Step3: Display the values of those variables.

Step4: Compute a = a+b

Step5: Display the addition of the two variables assigned to a.

Step6: Compute a = a-b

Step7: Display the difference of the two variables assigned to a.

Step8: Compute a = a*b

Step9: Display the multiplication of the two variables assigned to a.

Step10: Compute a = a/b

Step11: Display the division of a by b assigned to a.

Step12: Compute a = a%b

Step13: Display the remainder of the division of a by b assigned to a.

Step14: Stop.

PROGRAM 1:TO IMPLEMENT ASSIGNMENT OPERATORS

```
#include <stdio.h>
#include <conio.h>
intmain()
{
int a = 16;
int b = 5;
printf("\n a = %d  b = %d",a,b);
   a += b;
printf("\n %d",a);
   a -=b;
printf("\n%d",a);
   a *=b;
printf("\n%d",a);
   a /=b;
printf("\n%d",a);
   a %=b;
printf("\n%d",a);
getch();
}
```

OUTPUT:

ALGORITHM 2: INCREMENT AND DECREMENT OPERATOR

Step1: Start.

Step2: Declare the variables of type integers and assign the values.

Step3: Display the values of those variables.

Step4: Perform post increment operation – a++.

Step5: Display a – prints the value of a before increment.

Step6: Display a - prints the value of a after increment.

Step7: Perform pre increment operation – ++b.

Step8: Display b – prints the value of b after increment.

Step9: Display b – prints the value of b after increment.

Step10: Perform post decrement operation – c--.

Step11: Display c – prints the value of c before decrement.

Step12: Display c - prints the value of c after decrement.

Step13: Perform pre decrement operation – --d.

Step14: Display d – prints the value of d after decrement.

Step15: Display d – prints the value of d after decrement.

Step16: Stop.

PROGRAM 2: TO IMPLEMENT INCREMENT AND DECREMENT OPERATORS

```
#include <stdio.h>
#include <conio.h>
void main()
{
int a = 15, b = 15, c = 15, d = 15;
printf(" a = %d b = %d c = %d d = %d\n", a, b, c, d);
printf("\nvalue of a++ = %d", a++);
printf("\nvalue of ++b = %d", ++b);
printf("\nvalue of c-- = %d", c--);
printf("\nvalue of --d = %d", --d);
getch();
}
```

OUTPUT:

```
D:\BC5\BIN\pro7_inc_decop.exe
 a = 15  b = 15   c = 15  d = 15

value of a++ = 15
value of ++b = 16
value of c-- = 15
value of --d = 14
```

ALGORITHM 3: SIZE OF OPERATOR

Step1: Start.

Step2: Display size of integer data.

Step3: Display size of float data.

Step4: Display size of character data.

Step5: Display size of double data.

Step6: Stop.

PROGRAM 3: TO IMPLEMENT SIZEOF OPERATOR

```
#include <stdio.h>
#include <conio.h>
void main()
{
printf("\nSize of Integer %d", sizeof(int));
printf("\nSize of Character %d", sizeof(char));
printf("\nSize of Float %d", sizeof(short));
printf("\nSize of long %d", sizeof(long));
getch();
}
```

OUTPUT:

```
D:\BC5\BIN\pro3_sizeop.exe

Size of Integer 4
Size of Character 1
Size of Float 2
Size of long 4
```

ALGORITHM 4: BITWISE OPERATORS

Step1: Start.

Step2: Declare the variables integer variables a,b and assign values.

Step3: Display a and b.

Step4: Display result of bit wise and (a&b) .

Step5: Display result of bit wise or (a|b).

Step6: Display the result of bit wise xor (a^b) .

Step7: Display the result of bit wise complement (~a) .

Step8: Display the result of bit wise right shift by 2 (a>>2) .

Step9: Display the result of bit wise left shift by 2 (a<<2) .

Step10: Stop.

PROGRAM 4: TO IMPLEMENT BITWISE OPERATORS

```
#include <stdio.h>
#include <conio.h>
void main()
{
int a = 7;
int b = 11;
printf("\n a = %d    b = %d \n\n", a,b);
printf("\nBitwise AND (&) operator\n");
printf("%d\n",a&b);
printf("\nBitwise OR (|) operator\n");
printf("%d\n",a|b);
printf("\nBitwise XOR (^) operator\n");
printf("%d\n",a^b);
printf("\nBitwise COMPLEMENT (~) operator\n");
printf("%d\n",~a);
printf("\nBitwise RIGHT SHIFT (>>) operator\n");
printf("%d\n",a>>2);
printf("\nBitwise LEFT SHIFT (<<) operator\n");
printf("%d\n",a<<2);
getch();
}
```

OUTPUT:

```
D:\BC5\BIN\pro8_bitwiseop.exe

 a = 7    b = 11


Bitwise AND (&) operator
3

Bitwise OR (|) operator
15

Bitwise XOR (^) operator
12

Bitwise COMPLEMENT (~) operator
-8

Bitwise RIGHT SHIFT (>>) operator
1

Bitwise LEFT SHIFT (<<) operator
28
```

RESULT:

Assignment, increment and decrement operators, size of and bitwise operators are implemented using C programs.

Expt. No: **Date:**

8. PROGRAM TO FIND AREA AND VOLUME OF GEOMETRICAL SHAPES

AIM:

To write a program to find the areas and volumes of various geometrical shapes.

ALGORTIHM :

Step1: Start the program

Step2: Declare the integer variables a, b, h, r, area and volume .

Step3: Declare the float variables area1, vol and pi.

Step4:Assign 3.144 to pi.

Step5: Read the value for side of a square.

Step6: Compute area = a * a

Step7: Compute volume = a * a * a

Step8: Print the area of a square.

Step9: Print the volume of a square.

Step10: Read the values for length, breadth and height of a rectangle.

Step11: Compute area = a * b

Step12: Compute volume = a * b * b *h

Step13: Print the area of a rectangle.

Step14: Print the volume of a rectangle.

Step15: Read the value for radius of a circle.

Step16: Compute area1 = pi *r *r

Step17: Compute vol = pi * r * r * r

Step18: Print the area of a circle.

Step19: Print the volume of a circle.

Step20: End

PROGRAM:

```
#include <stdio.h>
#include <conio.h>
void main()
{
int a, b, h, r, area, volume;
float area1, vol,pi=3.144;
printf(" \n Enter the side of a square :");
scanf("%d",&a);
area = a*a;
volume = a*a*a;
printf("\n Area of a square = %d",area);
printf("\n Volume of a square = %d",volume);
printf("\n\n Enter length, breadth and height of a rectangle : ");
scanf("%d %d %d",&a,&b, &h);
area = a*b;
volume = a*b*h;
```

```
printf("\n Area of a rectangle = %d ", area);

printf("\n Volume of a rectangle = %d",volume);

printf("\n\n Enter the radius of a circle: ");

scanf("%d",&r);

area1 = pi*r*r;

vol = pi*r*r*r;

printf("\n\n Area of circle = %f ",area1);

printf("\n Volume of a circle = %f",vol);

getch();

}
```

OUTPUT:

D:\BC5\BIN\pro8_areaofshapes.exe

```
Enter the side of a square :4

Area of a square = 16
Volume of a square = 64

Enter length, breadth and height of a rectangle : 3 4 5

Area of a rectangle = 12
Volume of a rectangle = 60

Enter the radius of a circle: 6


Area of circle = 113.184006
Volume of a circle = 679.104004
```

RESULT:

Area and volume of various geometrical shapes are obtained by using a C program.

Expt. No: **Date:**

9. PROGRAM TO FIND THE BIGGEST OF THREE NUMBERS

AIM:

To write a program to find the biggest of three numbers.

ALGORITHM :

Step 1: Start

Step 2: Declare variables a,b and c.

Step 3: Read variables a,b and c.

Step 4: If a>b

If a>c

Display a is the largest number.

Else

Display c is the largest number.

Else

If b>c

Display b is the largest number.

Else

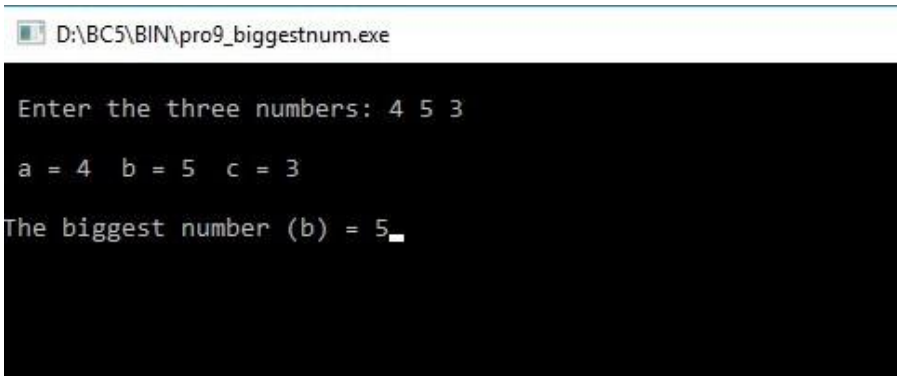
Display c is the greatest number.

Step 5: Stop

PROGRAM:

```
#include <stdio.h>
#include <conio.h>
void main()
{
int a, b, c;
printf("\n Enter the three numbers: ");
scanf("%d %d %d",  &a,  &b,  &c);
printf("\n a = %d b = %d c = %d", a, b, c);
if(a>b)
    {
        if(a>c)
            printf("\n\nThe biggest number (a) = %d", a);
        else
            printf("\n\nThe biggest number (c) = %d", c);
    }
else
    {
        if(b>c)
            printf("\n\nThe biggest number (b) = %d", b);
        else
            printf("\n\nThe biggest number (c) = %d", c);
    }
getch();
}
```

OUTPUT:


D:\BC5\BIN\pro9_biggestnum.exe
Enter the three numbers: 4 5 3
a = 4 b = 5 c = 3
The biggest number (b) = 5

USING IF.. ELSE IF .. ELSE TO FIND GREATEST NUMBER

ALGORITHM:

Step 1: Start

Step 2: Declare variables a,b and c.

Step 3: Read variables a,b and c.

Step 4: If a>b and a>c then

Display a is the largest number.

Else if b>a and b>c then

Display b is the largest number.

Else

Display c is the largest number.

Step 5: Stop

PROGRAM:

```
#include <stdio.h>
#include <conio.h>
void main()
{
int a, b, c;
printf("Enter three numbers\n");
scanf("%d %d %d",&a,&b,&c);
if(a>b && a>c)
  {
      printf("\n %d is greatest\n",a);
```

```
    }
    else if(b>a && b>c)
    {
        printf("\n %d is greatest\n",b);
    }
    else
    {
         printf("\n %d is greatest\n",c);
    }
getch();
}
```

OUTPUT:

```
D:\BC5\BIN\pro9_biggestnum1.exe
Enter three numbers
8 5 7

 8 is greatest
```

RESULT:

The biggest number from the given three numbers is found using C programs.

Expt. No: **Date:**

10. PROGRAM TO PRINT THE MULTIPLICATION TABLE

AIM:

To write a program to print the multiplication table.

ALGORITHM:

Step 1: Start

Step 2: Declare variables mult_term, repet_term.

Step 3: Read the value for mult_term.

Step 4: Initialize repet_term = 1.

Step 5: Repeat the steps until repet_term≤10.

5.1: Display repet_term x mult_term = (repet_term * mult_term)

Step 6: Stop

PROGRAM:

```
#include <stdio.h>
#include <conio.h>
void main()
{
inta ,i ;
printf("\nEnter which Multiplication table to be printed:" );
scanf("%d",&a);
for (i=1; i<=10; i++)
    printf("\n %d x %d = %d", i,a,i*a);
getch();
}
```

OUTPUT:

```
D:\BC5\BIN\pro10_multable.exe

Enter which Multiplication table to be printed:6

 1   x   6  =   6
 2   x   6  =   12
 3   x   6  =   18
 4   x   6  =   24
 5   x   6  =   30
 6   x   6  =   36
 7   x   6  =   42
 8   x   6  =   48
 9   x   6  =   54
 10   x   6  =   60
```

RESULT:

Multiplication table is printed by using a C program.